LP Duality – Lower Bounds – IP (Lec12)

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Todays topics & Logistics

- Forming the dual LP of a general primal LP
- Economic interpretation of dual variables
- Example

Recap

Duality in LP

- **Weak Duality:** if x is feasible for the Primal problem and y is feasible for the Dual problem, then $c^T x \leq b^T y$.
  - That is, any feasible solution of the Dual problem is an Upper Bound of the Primal problem. The best (smaller) Upper Bound is the optimal solution of the Dual problem
  - **Proof**: $b^T y = y^T b \geq y^T A x = x^T A^T y \geq c^T y$ Hence: $b^T y \geq c^T x$
- **Strong Duality:** if the Primal problem has an <u>optimal solution</u> x^* with a finite optimal value ($f(x^*) < \infty$), then the Dual problem also has an <u>optimal value</u> $\boldsymbol{y}^*$ with a finite optimal value ($g(y^*) < \infty$), and $c^T x^* = b^T y^*$
- **Duality gap:** $\mathrm{d} = b^T y^* - c^T x^*$.

Primal problem

$$\begin{aligned} max \quad & f(x) = c^T x \\ s.t. \quad & Ax \leq b \\ & \mathrm{x} \geq 0 \end{aligned}$$

Dual problem

$$\begin{aligned} min \quad & g(y) = b^T y \\ s.t. \quad & A^T y \geq c \\ & \mathrm{y} \geq 0 \end{aligned}$$

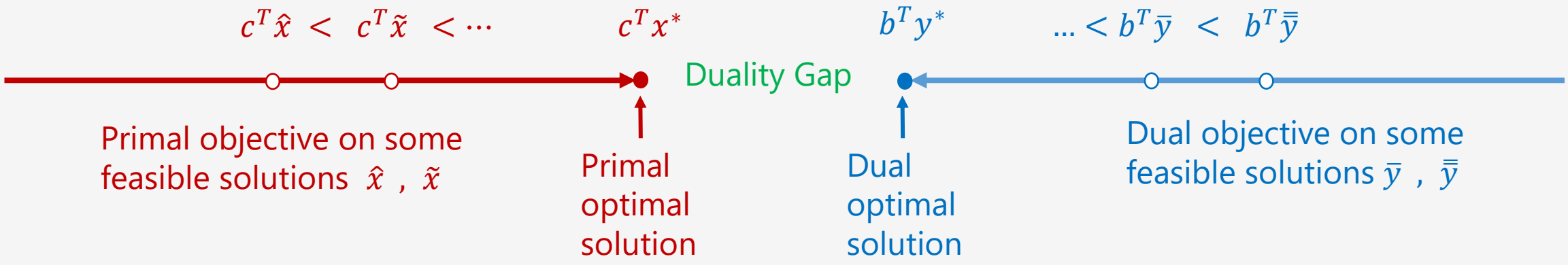
Graphical representation of Primal & Dual problems

Primal problem

$$\begin{aligned} max \quad & f(x) = c^T x \\ s.t. \quad & Ax \le b \\ & x \ge 0 \end{aligned}$$

Dual problem

$$\begin{aligned} min \quad & g(y) = b^T y \\ s.t. \quad & A^T y \ge c \\ & y \ge 0 \end{aligned}$$

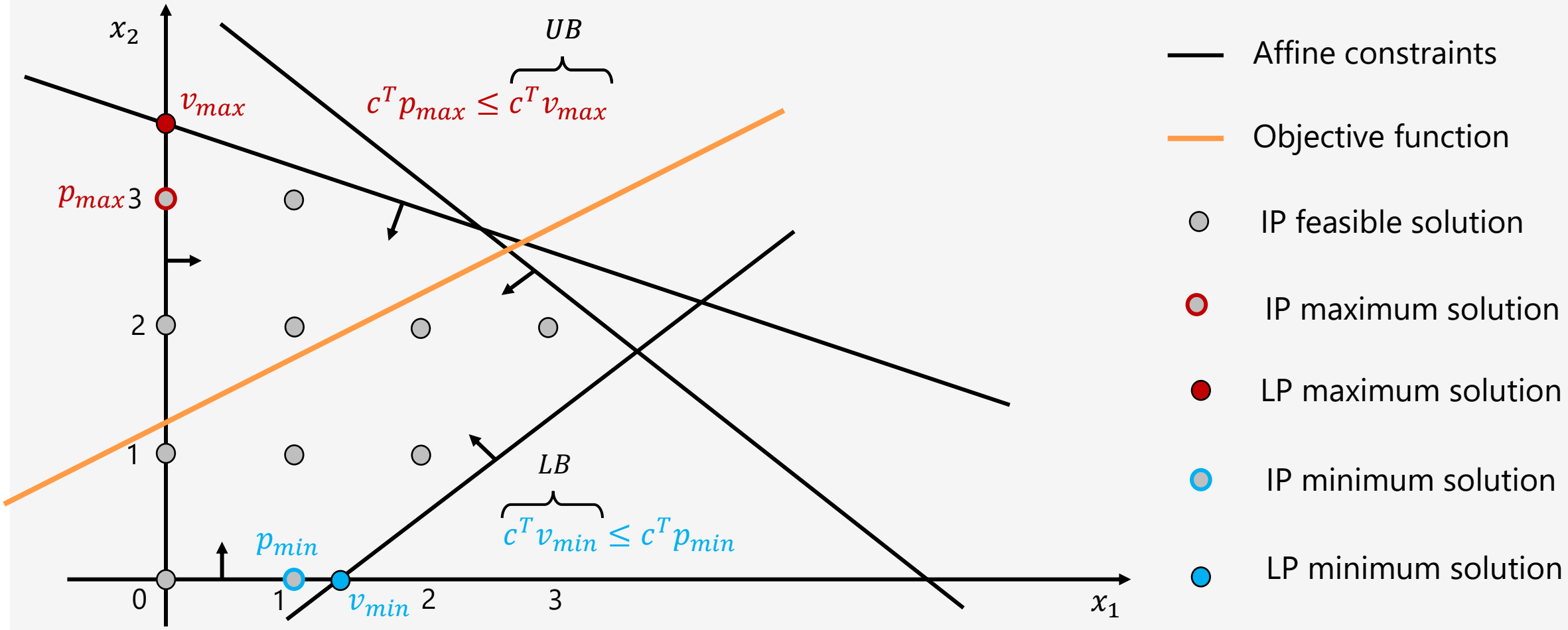


Weak duality: $c^T\tilde{x} < b^T\bar{y}$

Strong duality: $c^T x^* = b^T y^*$

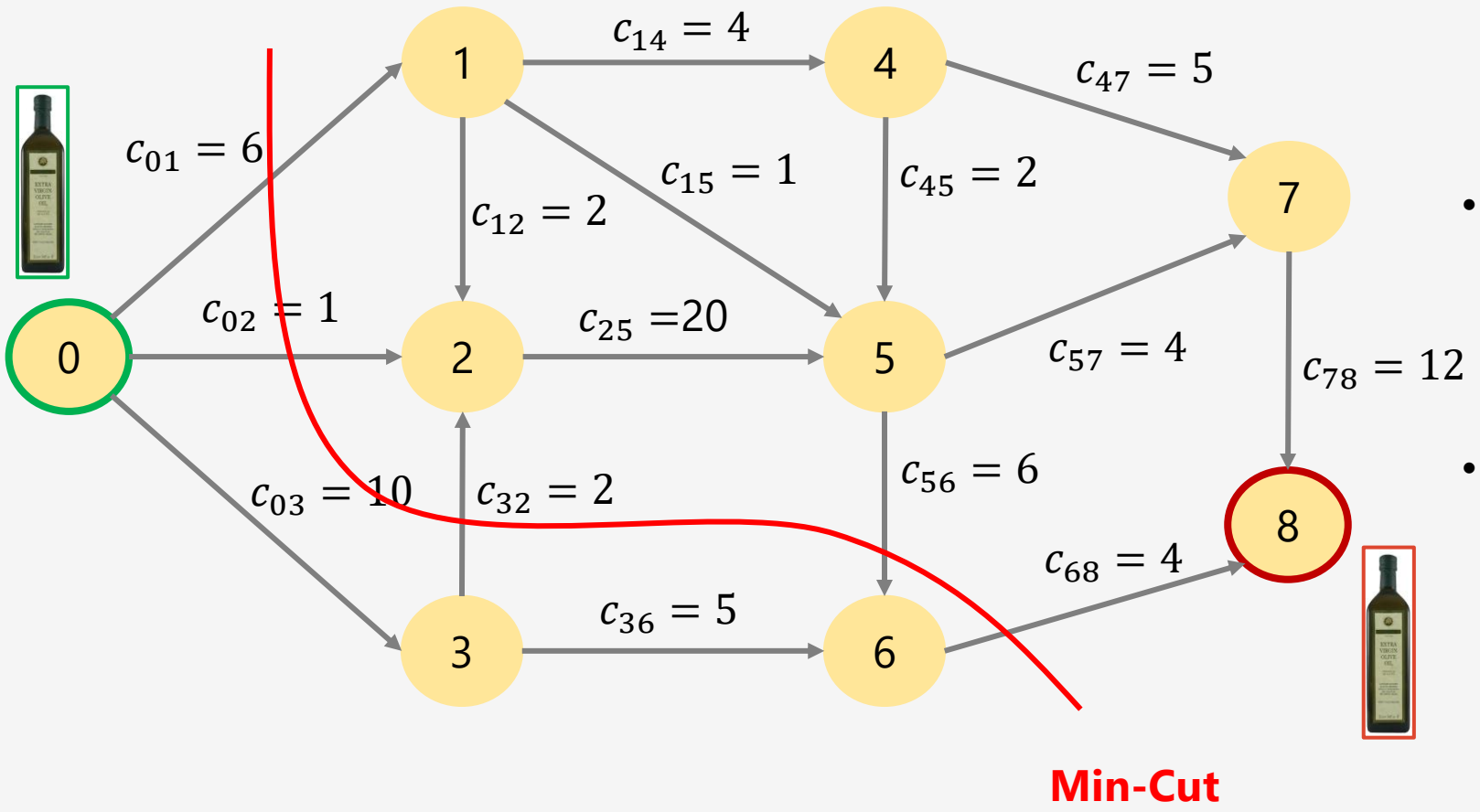
Duality theory helps us **certifying optimality** of the primal problem or to find an **upper bound** of it.

Lower and Upper Bounds based on **LP relaxations** of **IP problems**



		PRIMAL		
		Finite optimal	Unbounded	Infeasible
DUAL	Finite optimal	Possible	Impossible	Impossible
	Unbounded	Impossible	Impossible	Possible
	Infeasible	Impossible	Possible	Possible

Max-Flow and Min-Cut are dual of each other



- **Decision variables:** quantity of olive oil transported from node i to node j.
  - x_{ij}
- **Objective**: Ship as much olive oil as possible from the source node 0 to the terminal node 8.
  - $\max f(x) = x_{01} + x_{02} + x_{03}$
- **Constraints:**
  - Cannot exceed the capacity of the edges
    - $0 \leq x_{ij} \leq c_{ij}$
  - flowIN == flowOUT
    - $x_{01} = x_{14} + x_{15} + x_{12}$
    - $x_{14} = x_{47} + x_{45}$
    - $x_{15} + x_{45} + x_{25} = x_{57} + x_{56}$
    - …

Forming the Dual LP of a general primal LP

Converting min to max and vice versa

- Converting a minimization LP to a maximization LP

$$\begin{aligned} min \quad & c^T x \\ s.t. \quad & Ax \le b \\ & x \ge 0 \end{aligned} \qquad\qquad -\begin{aligned} max \quad & -c^T x \\ s.t. \quad & Ax \le b \\ & x \ge 0 \end{aligned}$$

- Converting a maximization LP to a minimization LP

$$\begin{aligned} max \quad & c^T x \\ s.t. \quad & Ax \le b \\ & x \ge 0 \end{aligned} \qquad\qquad -\begin{aligned} min \quad & -c^T x \\ s.t. \quad & Ax \le b \\ & x \ge 0 \end{aligned}$$

- **Note**: we multiply the objective **function** by -1 **and** the objective function **value** by -1.

- The two LPs have the **same** optimal solution $\boldsymbol{x}^*$ <u>**and**</u> the **same** optimal objective function value, $\boldsymbol{c^T x^*}$

- If you **do not** multiply the objective function value by -1, the two problems <u>still have the same optimal solution</u> $\boldsymbol{x}^*$ **but** the <u>objective function values</u> have <u>opposite sign</u>

Deriving the Dual LP from different formulations

- Consider the primal LP:

Primal problem

$$\begin{aligned} min \quad & c^T x \\ s.t. \quad & Ax \le b \\ & x \ge 0 \end{aligned}$$

- Easy to remember how to find its dual LP:

Dual problem

$$\begin{aligned} max \quad & g(y) = b^T y \\ s.t. \quad & A^T y \ge c \\ & y \ge 0 \end{aligned}$$

- There is helpful **symmetry**

- Now consider the primal LP:

Primal problem

$$\begin{aligned} max \quad & c^T x \\ s.t. \quad & Ax \le b \\ & x \ge 0 \end{aligned}$$

$\Rightarrow$

Primal problem

$$\begin{aligned} -\,\mathbf{min} \quad & -c^T x \\ s.t. \quad & Ax \le b \\ & x \ge 0 \end{aligned}$$

- We can use the easy to remember conversion:

Dual problem

$$\begin{aligned} -\,max \quad & b^T y \\ s.t. \quad & A^T y \ge -c \\ & y \ge 0 \end{aligned}$$

$\Rightarrow$

Dual problem

$$\begin{aligned} min \quad & -b^T y \\ s.t. \quad & A^T y \ge -c \\ & y \ge 0 \end{aligned}$$

Deriving the Dual LP from different formulations

- Consider the primal LP (with equality constraints):

Primal problem

$$\begin{aligned} min \quad & c^T x \\ s.t. \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

We try to convert it to the symmetric form of the previous slide!

Primal problem

$$\begin{aligned} min \quad & c^T x \\ s.t. \quad & Ax \leq b \\ & Ax \geq b \\ & x \geq 0 \end{aligned}$$

$\Rightarrow$

Primal problem

$$\begin{aligned} min \quad & c^T x \\ s.t. \quad & \begin{bmatrix} A \\ -A \end{bmatrix} x \leq \begin{bmatrix} b \\ -b \end{bmatrix} \\ & x \geq 0 \end{aligned}$$

$\Rightarrow$

Dual problem

$$\begin{aligned} max \quad & b^T y_1 - b^T y_2 \\ s.t. \quad & A^T y_1 - A^T y_2 \geq c \\ & y_1, y_2 \geq 0 \end{aligned}$$

$\Rightarrow$

Dual problem

$$\begin{aligned} max \quad & b^T (y_1 - y_2) \\ s.t. \quad & A^T (y_1 - y_2) \geq c \\ & y_1, y_2 \geq 0 \end{aligned}$$

Note that every real vector can be written as a difference of two positive vectors.

Hence define: $y = y_1 - y_2$ with $y \in R^m$ free variable

Dual problem

$$\begin{aligned} max \quad & b^T y \\ s.t. \quad & A^T y \geq c \end{aligned}$$

Deriving the Dual LP from different formulations

- **Exercises**:

- Derive the dual of the LP:

Primal problem

$$\begin{aligned} \min \quad & c^T x \\ s.t. \quad & Ax = b \end{aligned}$$

Dual problem

$$\begin{aligned} \max \quad & b^T y \\ s.t. \quad & A^T y = c \end{aligned}$$

- Derive the dual of the LP:

Primal problem

$$\begin{aligned} \min \quad & c^T x \\ s.t. \quad & \begin{aligned} Ax &\leq b \\ Dx &= f \end{aligned} \end{aligned}$$

Dual problem

$$\begin{aligned} \max \quad & b^T y + f^T z \\ s.t. \quad & \begin{aligned} A^T y + D^T z &= c \\ y \geq 0, z\ & free \end{aligned} \end{aligned}$$

Hint 1: since $x \in R^n$ is free variable, you can write it as $x = x_1 - x_2$ with $x_1, x_2 \geq 0$

Hint 2: you can convert the equality constraints to $\leq$ constraints we did in the previous slide.

After following the two hints you can apply the symmetric form for deriving the dual.

The Dual of the Dual is the Primal

Primal problem

$$\begin{aligned} min \quad & c^T x \\ s.t. \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

Dual problem

$$\begin{aligned} max \quad & b^T y \\ s.t. \quad & A^T y \geq c \\ & y \geq 0 \end{aligned}$$

Dual of Dual problem

$$\begin{aligned} min \quad & c^T x \\ s.t. \quad & Ax \leq b \\ & x \geq 0 \end{aligned}$$

This is true for any other formulation of an LP and its dual.

Interpretation of Dual LP and Dual Variables

Example: a small company produces: **tables** and **chairs**.

- A table requires 2 units of **wood** and 1 hour of **labor**
- A chair requires 2 unit of **wood** and 1 unit of **labor**
- The company's **resources**: 60 units wood and 50 units of labor
- Selling 1 table and 1 chair brings $40 and $30 of **profit**, respectively.

Primal LP

$$\begin{array}{ll} max & 40x_1 + 30x_2 \\ s.t. & 2x_1 + 2x_2 \leq 60 \\ & 1x_1 + 1x_2 \leq 50 \\ & x_1, x_2 \geq 0 \end{array}$$

Total profit
Wood constraint
Labor constraint

Interpretation: a **competitor** wants to buy your resources.

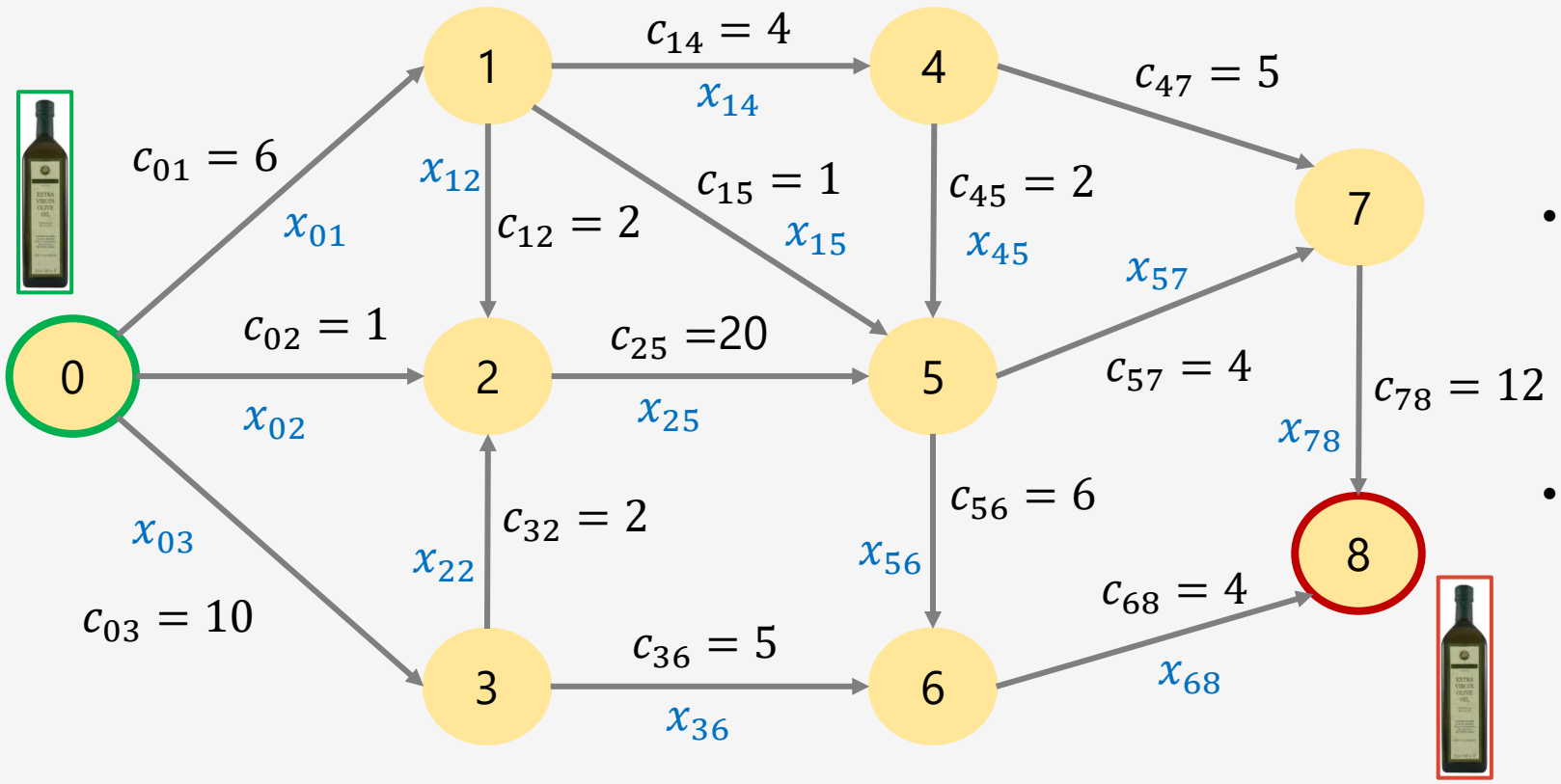
- The buyer must offer enough $ so you are **willing to sell your resources**.
- The buyer must offer $2y_1 + 1y_2 \geq 40$ to **buy** the resources for tables.
- Similarly, he must offer $2y_1 + 1y_2 \geq 30$ to **buy** the resources of the chairs.
- $\boldsymbol{y_1}$ and $\boldsymbol{y_2}$ represent the **value** of 1 unit of wood and 1 unit of labor
- $\boldsymbol{y_1}$ and $\boldsymbol{y_2}$ are called **shadow prices**

Dual LP

$$\begin{array}{ll} min & 60y_1 + 50y_2 \\ s.t. & 2y_1 + 1y_2 \geq 40 \\ & 2y_1 + 1y_2 \geq 30 \\ & y_1, y_2 \geq 0 \end{array}$$

Resources valuation
Table value
Chair value

Maximum Flow Problem



Graph $G = (V, E)$ $\quad V = \{0,1,2,\dots,8\}$ $\quad E = \{(0,1),(0,2),\dots(7,8)\}$

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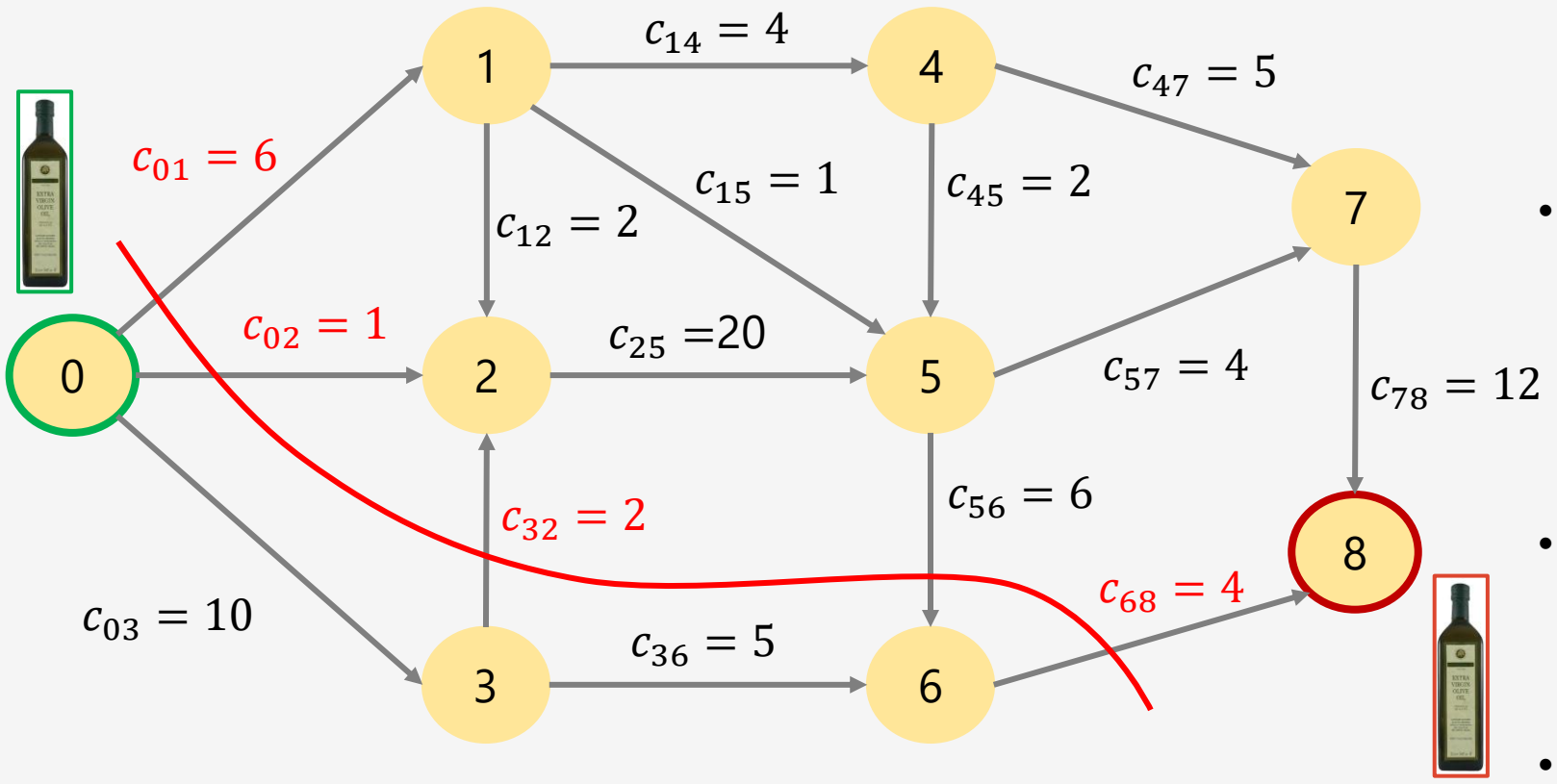
Minimum Cut Problem



Graph $G = (V, E)$ $\quad V = \{0,1,2,\dots,8\}$ $\quad E = \{(0,1),(0,2),\dots(7,8)\}$

- **What is a cut: a** partition of the nodes of graph in 2 disjoint subsets one containing the source and the other the terminal node
- **Objective:** find the cut with the minimum sum of capacities going from the subset that includes the source to the subsets that includes the terminal node.
- **Remark:** if we take the dual of the max-flow problem we will construct the min-cut problem!
- Strond and Weak Duality is perfectly demonstrated in this example.